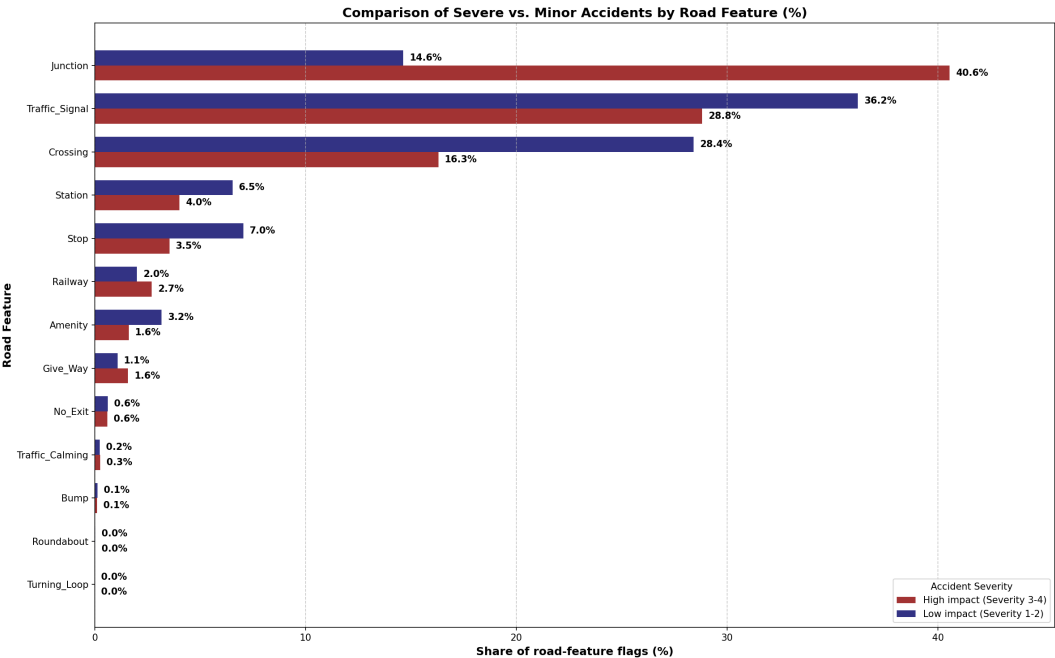


Summary Report: Which Road Features Are Linked to High-Impact Accidents?

Premiga Pengtap · Data: US Accidents (Kaggle, March 2023), 2016–2023

This report compares the mix of road features in high-impact accidents (Severity 3–4) with low-impact accidents (Severity 1–2). In this dataset, **Severity measures how much an accident disrupted traffic** (length of delay), not how many people were injured.



Road feature	High impact (3–4)	Low impact (1–2)	Difference
Junction	40.6%	14.6%	+26.0 pp
Railway	2.7%	2.0%	+0.7 pp
Give Way	1.6%	1.1%	+0.5 pp
Stop	3.5%	7.0%	–3.5 pp
Traffic Signal	28.8%	36.2%	–7.4 pp
Crossing	16.3%	28.4%	–12.1 pp

Key findings

- **Junctions are the clear standout.** They make up 40.6% of flags in high-impact accidents but only 14.6% in low-impact ones, almost three times as much.
- **Traffic signals, crossings and stop signs lean towards low-impact accidents.** Traffic signals are 28.8% of high-impact flags vs 36.2% of low-impact flags, and crossings 16.3% vs 28.4%. Controlled, slower-moving locations may produce more small collisions such as rear-end bumps.
- Railway crossings and give-way signs lean slightly towards high impact, but their numbers are small.

What this means

Traffic management and incident-response teams could give junctions priority for monitoring and fast clearance, since accidents there are more likely to cause long delays. Junction design reviews are a good starting point for reducing high-impact accidents.

Limitations

Severity reflects traffic delay, so these results should not be read as injury risk. Percentages are shares of road-feature flags within each group, not accident rates, and one accident can carry several flags.